

KIMStage

Control of a KIM101 piezo inertia stage in steps or micrometres.

Piezo inertia actuators move in steps whose size depends on the direction and the drive voltage. The module converts between steps and micrometres with step sizes entered by hand or measured with the camera module. It is in use on the lab setup. The window has a dark and a light theme.

ThorLabs KIM101 controller · three PIA25 inertia actuators (X, Y, Z)

Module source on GitHub

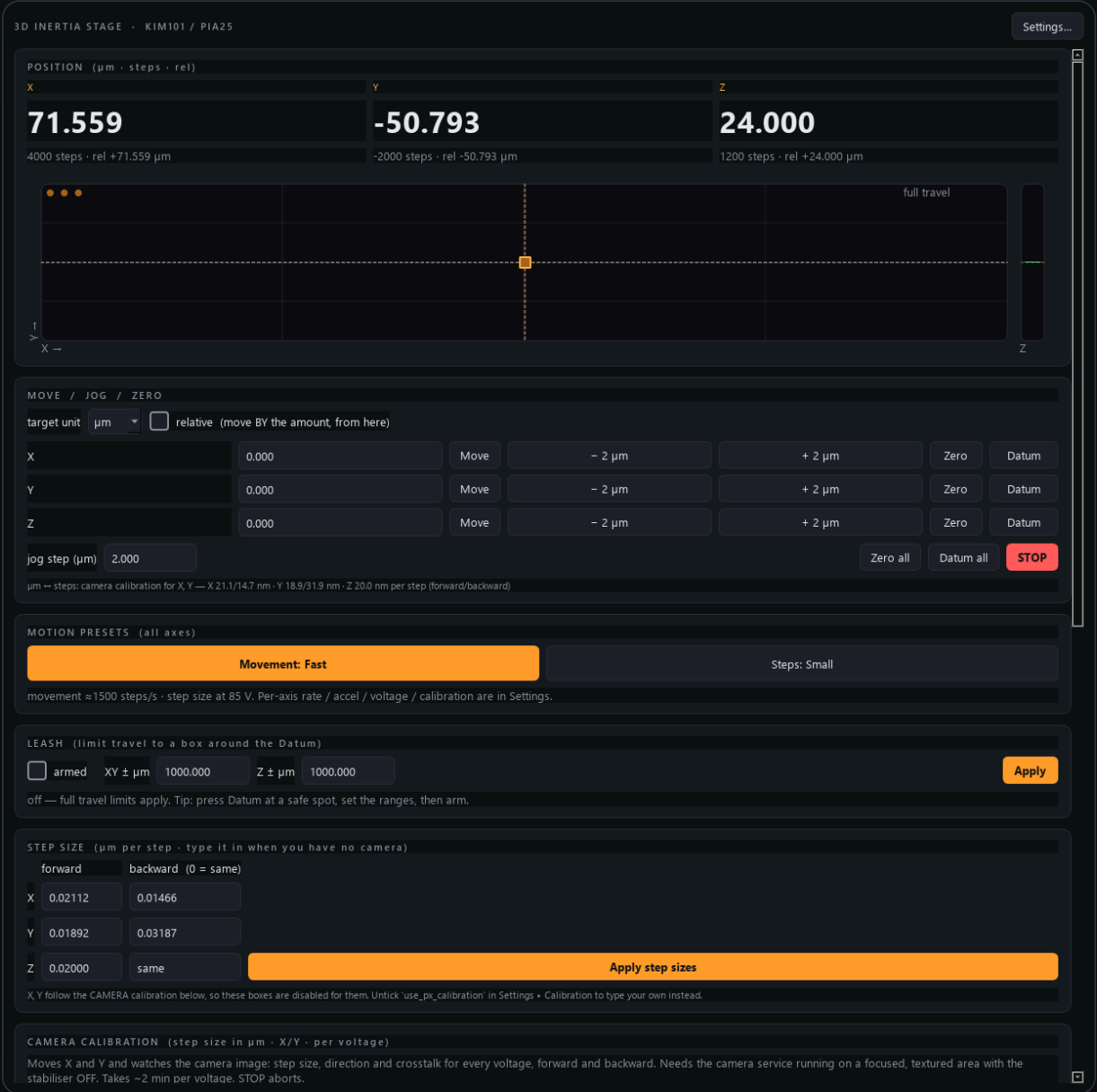
AaltoFlow on GitHub

µm or steps step-size calibration leash position list simulator included

MOVE

Step-size calibration and leash

A step forward and a step backward are not the same size on an inertia actuator (on the lab setup they differ by up to 70 %). The step size of each axis and direction can be entered by hand. Together with the camera module it can also be measured automatically: the stage moves, the camera measures the image shift, and the real step size is obtained for the positive and the negative direction of X and Y and for each drive voltage. Every micrometre move then uses the step size of its own direction. The leash is a box around the datum, set separately for XY and Z: while it is armed, no move can leave it.



Position, moves, presets and the leash (simulation)

Presets

Fast/slow speed and large/small step size; the defaults are the slow, small settings.

Steps or micrometres

Every move is available in both units.

Datum and display zero

The datum resets the step counters; the display zero only sets the origin of the relative readout.

Position list

Store, recall and save up to 20 positions.

Image moves

The camera module requests moves in image pixels; this module converts them to steps per direction.

Scans

X, Y and Z are scan axes in AaltoFlow; a scan waits until the stage has stopped.

PART OF AALTOFLOW

AaltoFlow

In AaltoFlow each instrument runs as its own service with its own window. A launcher starts them, and a scan engine drives any combination of them through multi-dimensional measurements.

github.com/FlashLukas/AaltoFlow

[kim-control](#) on GitHub

LICENCE AND CITATION

Citation

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L. Flajšman, *AaltoFlow: lab automation built from independent instrument modules*, NanoSpin group, Aalto University, doi:10.5281/zenodo.22959230

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